

CM0859 – MT5009 Type Theory

Natural Deduction

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Preliminaries

Textbook

“Lecture Notes on Natural Deduction” (Pfenning 2023).

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Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Preliminaries

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if __, then __)	conditional
$\perp$	(falsity)	bottom, falsum
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

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Definition

We define negation by $\neg A := A \supset \perp$.

Formalisation of Logical

In relation to the notions of **propositions** and **proofs** the textbook says:

“There is no universal agreement about the proper foundations for these notions.” (p. L2.1)

Formalisation of Logical Systems

Logical System {
 Axiomatic proof theory (Hilbert)
 Structural proof theory (Gentzen) {
 Natural deduction
 Sequent calculus

Formalisation of Logical Systems[†]

Logical System {
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[†]Complementary material: “Basic Proof Theory” (Troelstra and Schwichtenberg [1996] 2000) and “Structural Proof Theory” (Negri and von Plato [2001] 2008).

Judgements and Propositions

“The cornerstone of Martin-Löf’s foundation of logic is a clear separation of the notions of judgment and proposition.” (p. L2.1)

Judgements and Propositions

“He [Martin-Löf (1996) and Martin-Löf (1987)] reasons that to judge is to know and that an evident judgment is an object of knowledge. A proof is what makes a judgment evident. In logic, we make particular judgments such as ‘ A is a proposition’ or ‘ A is true’, presupposing in the latter case that A is already known to be a proposition. To know that ‘ A is a proposition’ means to know what counts as a verification of A , whereas to know that ‘ A is true’ means to know how to verify A .” (Pfenning and Davies 2001, p. 512)

Judgements and Propositions

Examples (judgements)

- “ A is a proposition”.
- “The proposition A is true”.
- “The proposition A is false”.
- “The proposition A is true at time t ” (temporal logic).
- “The proposition A is necessarily true” (modal logic).
- “ σ is a type” (programming languages).
- “ M is a combinator/program” (programming languages).
- “The combinator/program M has type σ ” (programming languages).

Judgements and Propositions

From the sharp separation of the notions of judgement and proposition by Martin-Löf, we know that inference rules operate on judgements and logical constants operate on propositions.

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Introduction

- Propositions: $A, B, C, \dots$
- Judgement: A_{prop} and A_{true}
- Form of the inference rules

$$\frac{J_1 \quad \dots \quad J_n}{J} \text{ rule name}$$

where J (conclusion) and $J_1, \dots, J_n$ (premises) are all judgements.

- Types of inference rules: Formation, introduction and elimination rules

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Inference rules for conjunction

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \wedge B \text{ prop}}$$

- Introduction rule

$$\frac{A \text{ true} \quad B \text{ true}}{A \wedge B \text{ true}} \wedge I$$

- Elimination rules

$$\frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1$$

$$\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2$$

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Inference rules for implication

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \supset B \text{ prop}}$$

- Introduction rule[†]

$$\frac{\frac{}{A \text{ true}}^u \quad \vdots \quad \frac{B \text{ true}}{A \supset B \text{ true}}}{A \supset B \text{ true}} \supset I^u$$

- Elimination rule

$$\frac{A \supset B \text{ true} \quad A \text{ true}}{B \text{ true}} \supset E$$

[†]In the application of the $\supset I$ rule, we may discharge zero, one, or more occurrences of the assumption.

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Example

A derivation where the assumption and the conclusion are the same and the assumption is discharged once. A proof that $A \supset A$ true.

$$\frac{\frac{}{A \text{ true}}}{A \supset A \text{ true}} \supset I^u$$

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Example

A derivation using the same assumption twice. A proof that $(A \wedge B) \supset (B \wedge A)$ true.

$$\frac{\frac{\frac{}{A \wedge B \text{ true}}^u}{B \text{ true}} \wedge E_2 \quad \frac{\frac{}{A \wedge B \text{ true}}^u}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I \quad \frac{}{(A \wedge B) \supset (B \wedge A) \text{ true}} \supset I^u$$

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Example

A derivation where there is a vacuous discharge when using the inference rule $\supset I$. A proof that $A \supset (B \supset A)$ true.

$$\frac{\frac{\frac{\text{—————}^u}{A \text{ true}} \supset I \text{ (vacuous discharge of } A)}{B \supset A \text{ true}} \supset I^u}{A \supset (B \supset A) \text{ true}}$$

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Notation

Let Γ be a set of judgements and let A be a judgement. The relation $\Gamma \vdash A$ means that there is a derivation with conclusion A from the set of hypotheses Γ .

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Example

A derivation using the same hypothesis twice. A proof that $A \wedge B \text{ true} \vdash B \wedge A \text{ true}$.

$$\frac{\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2 \quad \frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I$$

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Examples

See many more examples in (Carr 2018).

References

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References

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